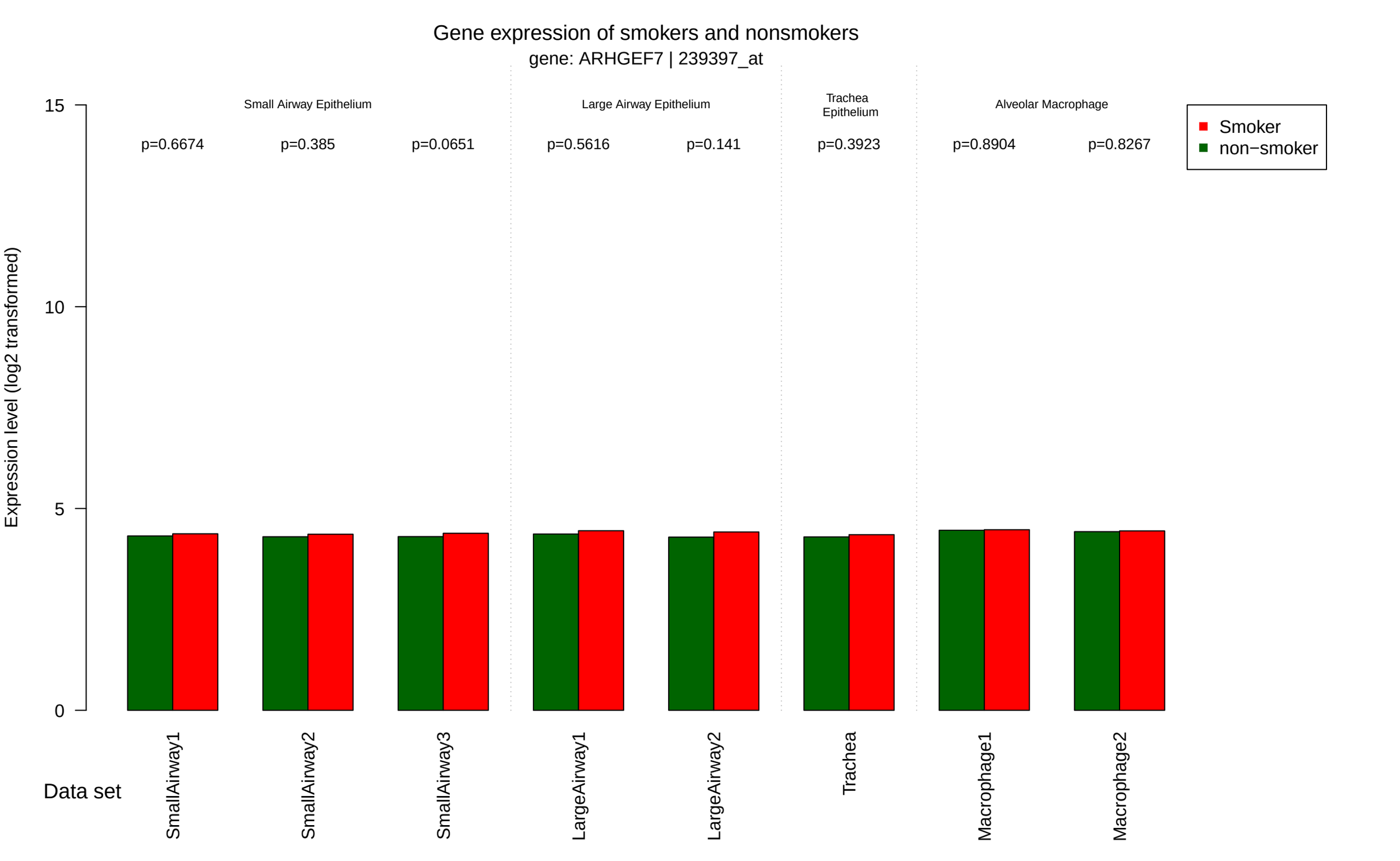
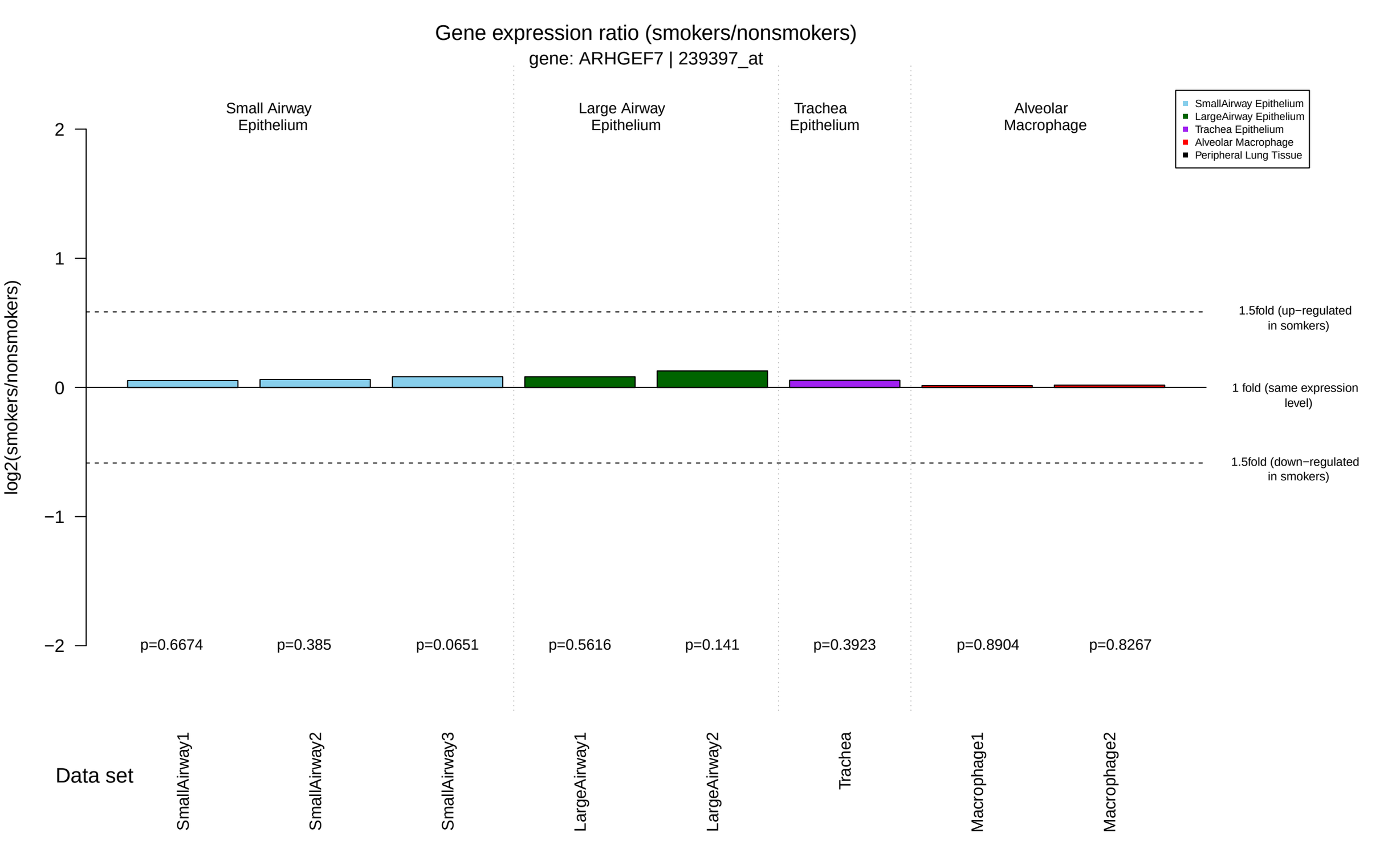
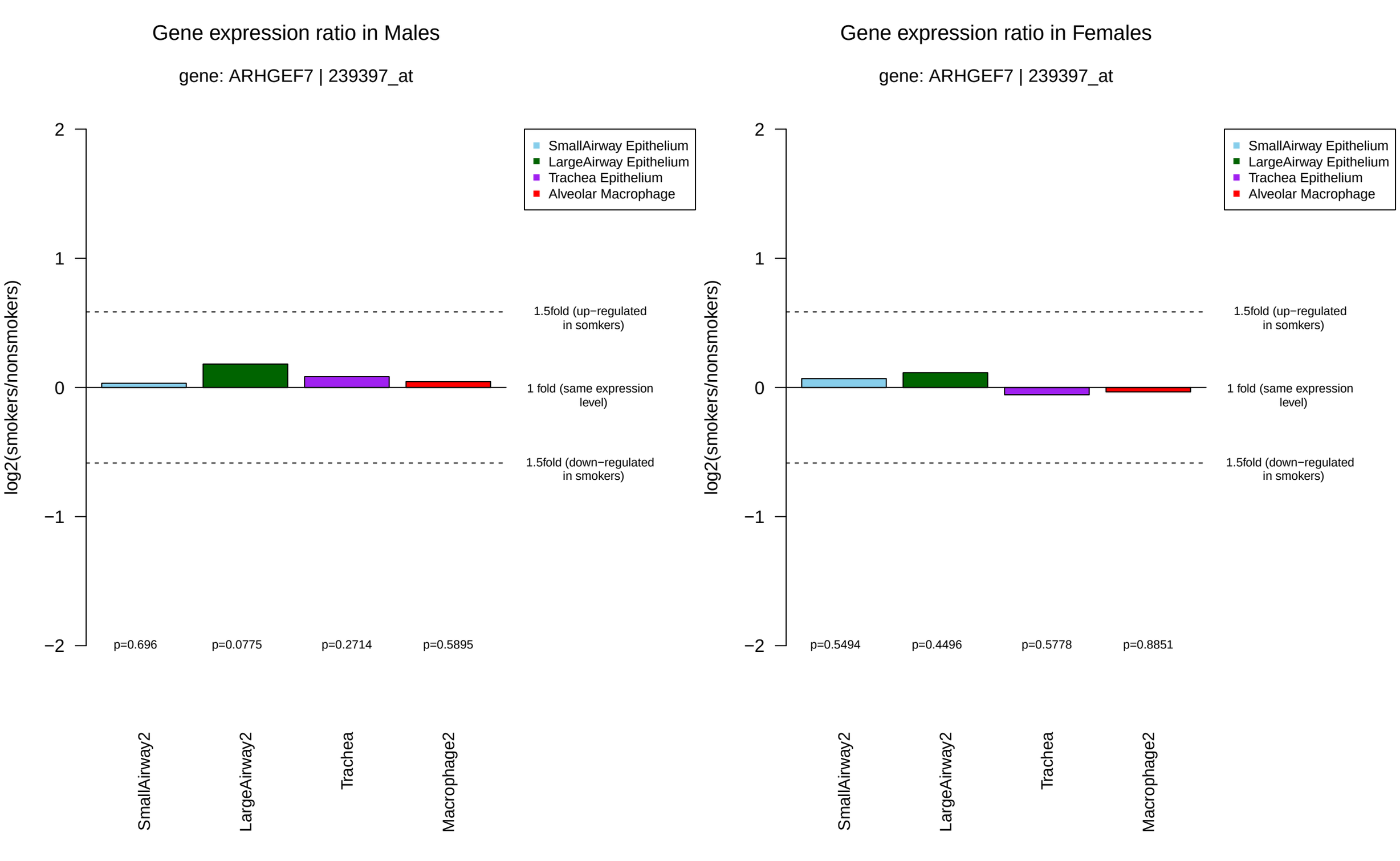
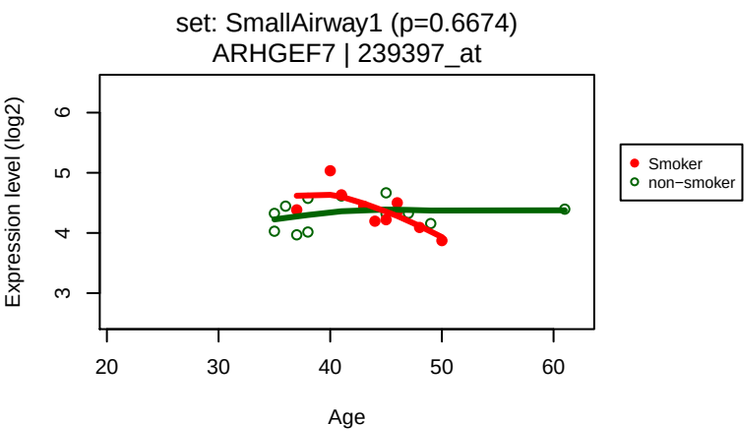
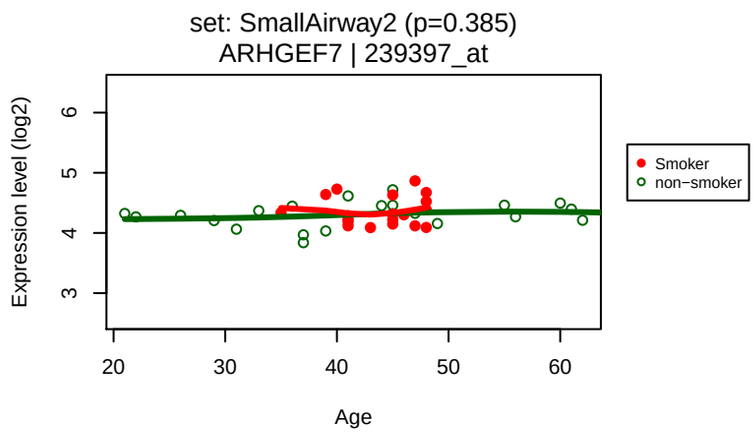
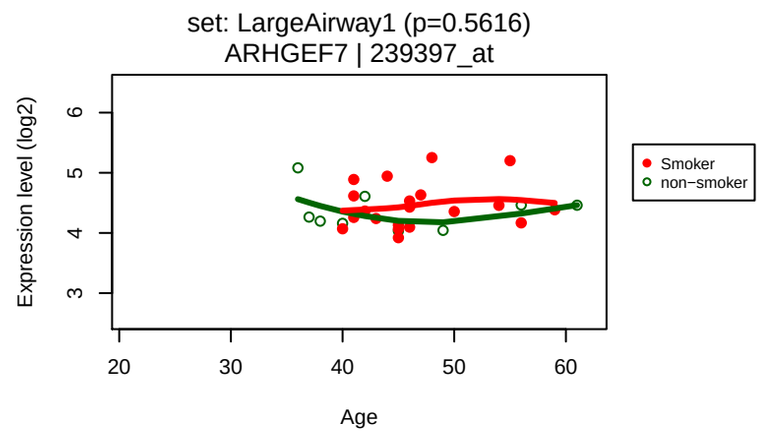

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 239397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6674
p=0.385
p=0.0651
p=0.5616
p=0.141
p=0.3923
p=0.8904
p=0.8267
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

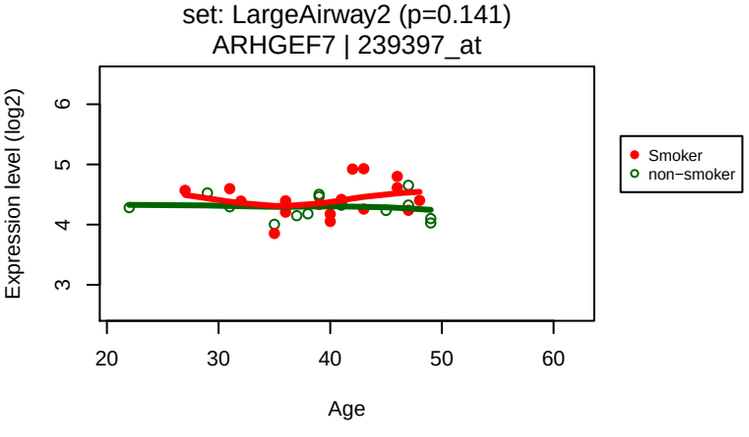

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 239397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6674
p=0.385
p=0.0651
p=0.5616
p=0.141
p=0.3923
p=0.8904
p=0.8267
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

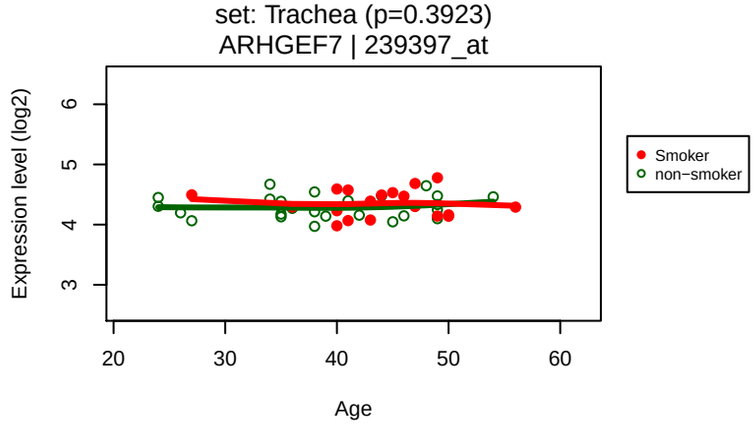

Gene expression ratio in Males
gene: ARHGEF7 | 239397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.696
p=0.0775
p=0.2714
p=0.5895
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 239397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5494
p=0.4496
p=0.5778
p=0.8851
SmallAirway2
LargeAirway2
Trachea
Macrophage2

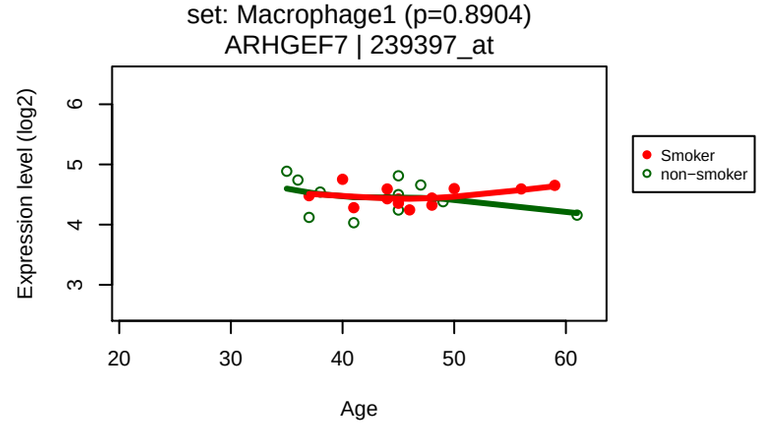

set: SmallAirway1 (p=0.6674)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker

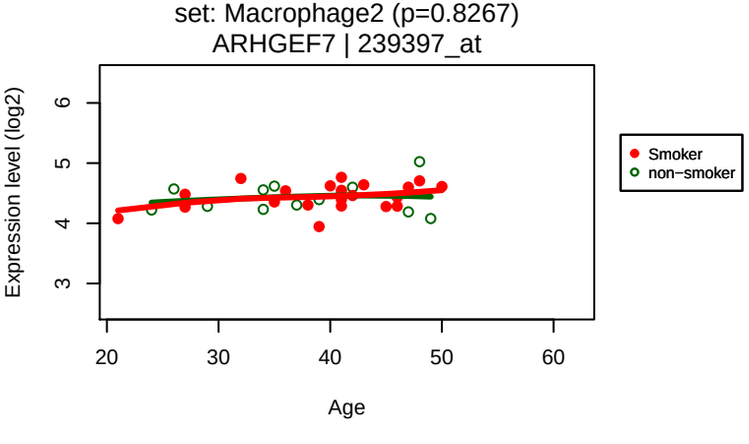

set: SmallAirway2 (p=0.385)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker

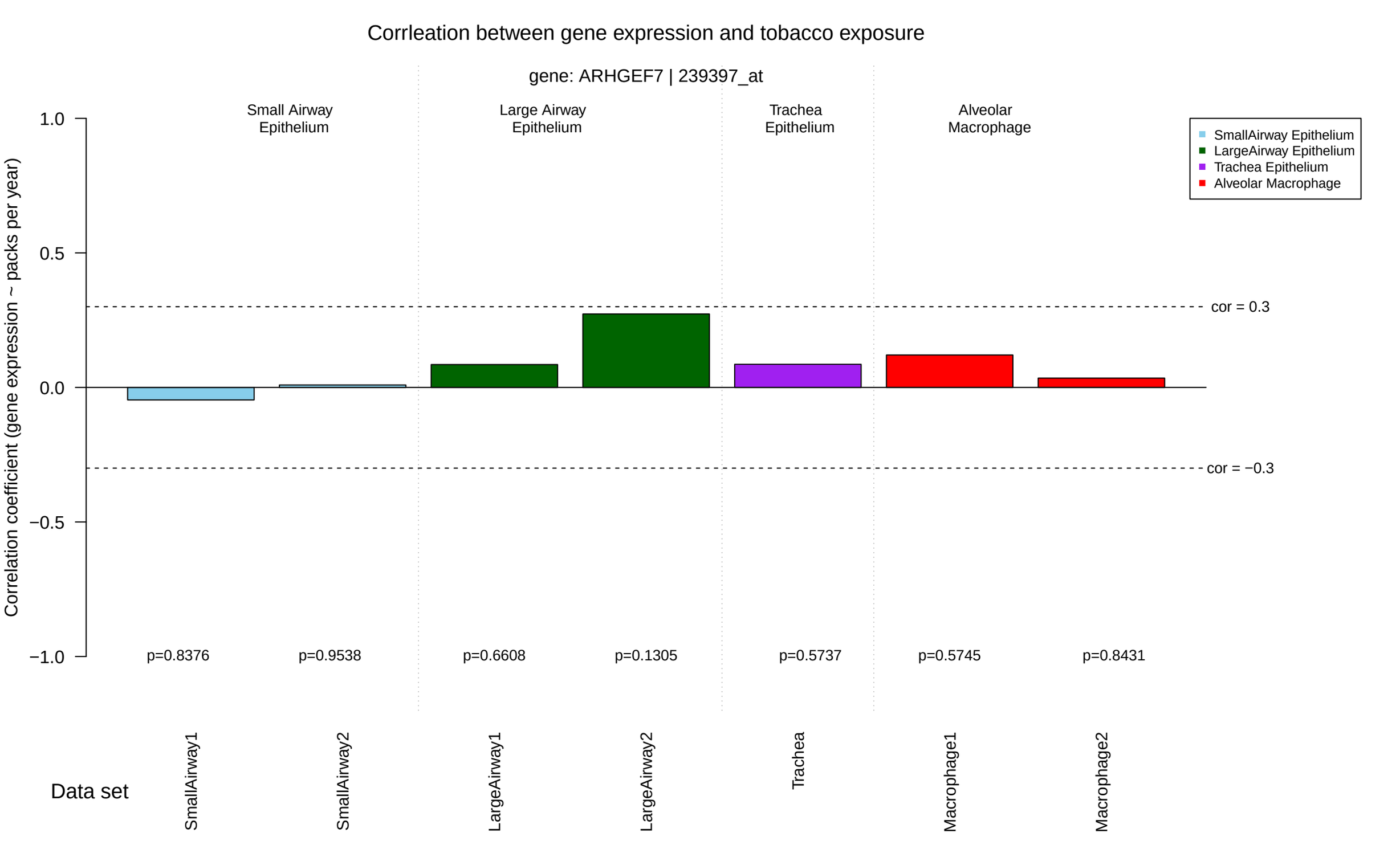

set: LargeAirway1 (p=0.5616)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.141)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3923)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8904)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8267)
ARHGEF7 | 239397_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 239397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8376
p=0.9538
p=0.6608
p=0.1305
p=0.5737
p=0.5745
p=0.8431
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2